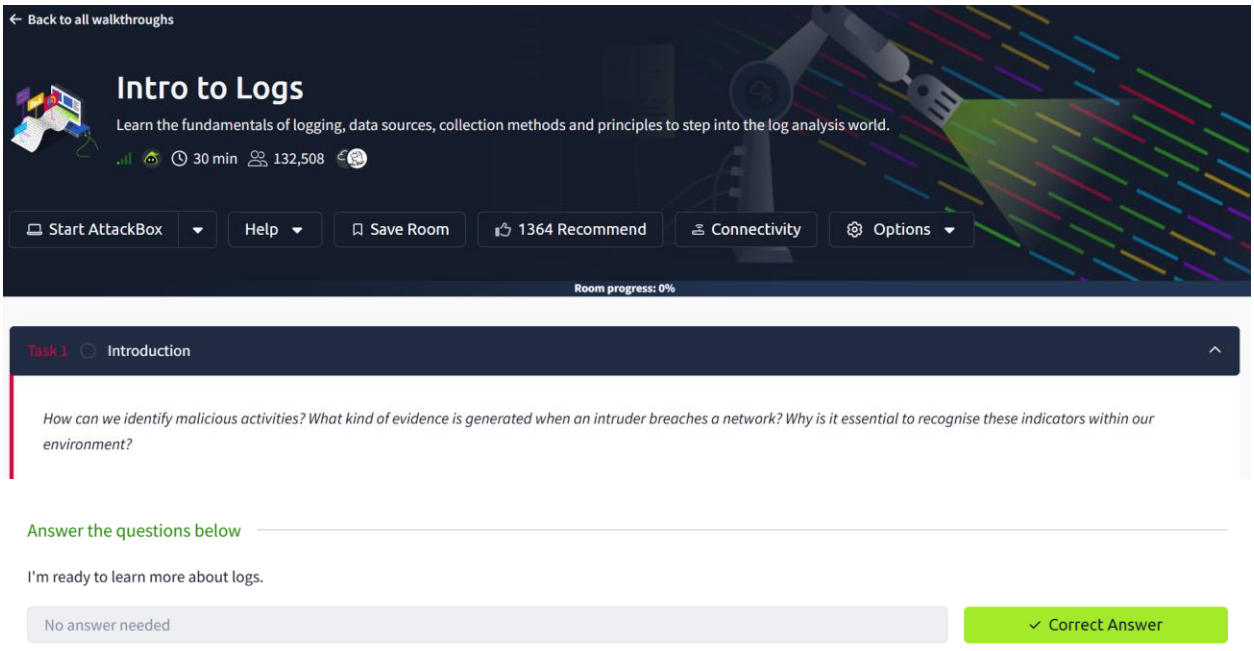
MARYAM REHAN

INTERNSHIP REPORT CYBER SECURITY (SOC)

Week 2 · Task 6 — Understanding Logs & Log Analysis Basics

Objective

The objective of this lab was to understand the importance of logs in cybersecurity, learn how log data records historical system and network activities, identify different log types and formats, and explore the processes involved in log collection, management, centralisation, storage, retention, and deletion. The lab also provided practical experience in analysing log files, identifying suspicious activities, and understanding how log analysis supports security monitoring and incident response.



Task 1: Introduction

Description

This task introduced the fundamental concepts of logging and explained how logs help maintain records of events occurring within a computer system or network. Logs provide information about system activities, user actions, application events, and potential security issues. Understanding these records is an important first step towards learning log analysis and security monitoring.

Purpose of Logs

Logs are created to maintain a historical record of system and application activities. They help administrators understand system operations, identify errors, investigate unexpected events, and review user activity. These records can also support investigations by providing timestamps and other details associated with specific events.

Importance of Logs in Cybersecurity

Logs play a significant role in detecting and investigating security incidents. Security analysts can examine authentication records, system events, and network activity to identify suspicious patterns. For example, repeated failed login attempts may indicate password guessing or a possible brute-force attack. Logs provide useful evidence, although suspicious activity must be investigated before drawing conclusions.

Role of Logs in Security Monitoring

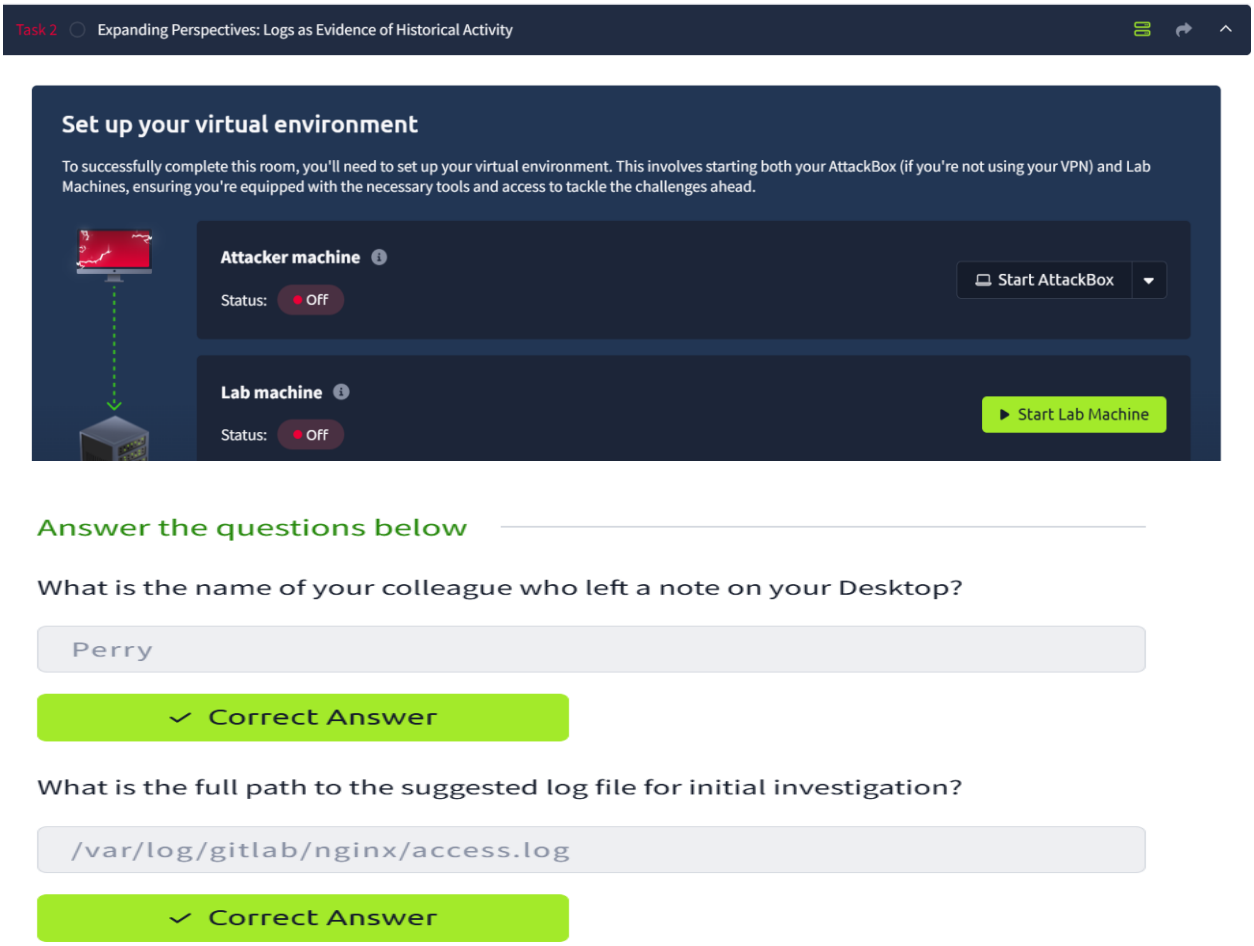
Security monitoring involves continuously reviewing system and network events to identify unusual or potentially harmful activities. Logs help security teams monitor authentication attempts, application behaviour, and changes to important system resources. In a Security Operations Center (SOC), log data can be analysed using monitoring tools and SIEM platforms to identify potential threats and support incident response.

Practical Learning Outcomes

After completing this introductory task, I gained a basic understanding of what logs are, why systems generate them, and how they support troubleshooting and cybersecurity investigations. This knowledge established a foundation for the next tasks, which explore log types, formats, collection, centralisation, storage, retention, and practical log analysis.

Conclusion

Task 1 established the basic understanding of logging and its importance in cybersecurity. I learned that logs preserve records of system activities and provide useful information for monitoring, troubleshooting, and investigating potential security incidents. These fundamental concepts are essential for understanding the more advanced log management and analysis techniques covered in the remaining tasks.



Task 2: Expanding Perspectives — Logs as Evidence of Historical Activity

Introduction

Logs are important sources of historical information because they maintain records of activities performed on computer systems, applications, and networks. They allow administrators and cybersecurity professionals to review past events and understand how a system was being used. Historical log analysis is particularly valuable when investigating suspicious activities or security incidents.

Task Description

This task focused on understanding how logs provide evidence of historical activity. Log records contain information about events that have occurred, including timestamps, user activities, system events, and application requests. By examining these records, security analysts can investigate previous activities and determine whether an event requires further attention.

The task highlighted that logs are not only useful for monitoring current system operations but also help reconstruct the sequence of events during an investigation.

Logs as Historical Evidence

Logs preserve records of activities that can be reviewed after an event has occurred. They help investigators establish a timeline and understand the order in which different actions took place. For example, authentication logs can show recorded login attempts, while web server logs can provide information about requests made to a website.

Historical evidence can help identify unusual behaviour, investigate unauthorised access attempts, and understand the possible causes of technical or security problems.

Importance of Timestamps

Timestamps indicate when a particular event was recorded. They help analysts arrange log entries chronologically and compare events from different sources. Accurate timestamps are important when building an incident timeline and determining the relationship between multiple activities.

However, analysts should also consider time zones, clock synchronisation, and the accuracy of timestamps when comparing records from different systems.

Identifying Suspicious Activities

Analysing historical logs can help security professionals recognise patterns that may indicate suspicious behaviour. Examples include repeated failed login attempts, unusual access requests, unexpected system changes, and activity outside normal operating patterns.

Such events do not automatically confirm a security breach. Analysts must examine the surrounding records and other available evidence before determining whether the activity is malicious.

Role of Historical Logs in Incident Response

Historical logs support incident response by helping investigators understand what happened before, during, and after a suspected security incident. They can assist in identifying affected systems, reviewing possible entry points, and determining which activities require additional investigation.

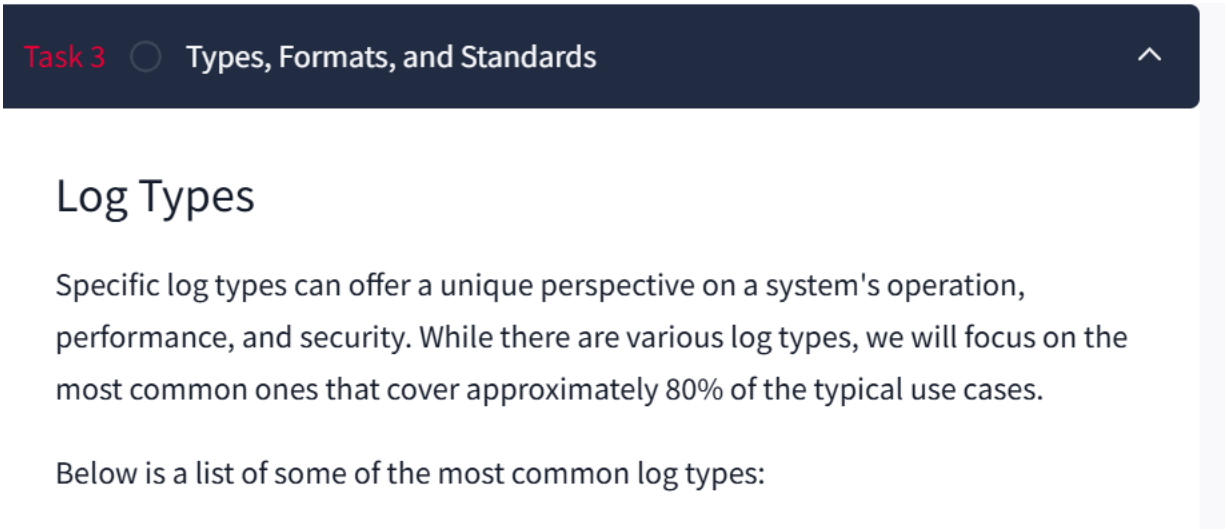
Preserving relevant logs is important because records may be rotated or deleted according to retention policies. Timely collection helps ensure that useful evidence remains available.

Learning Outcomes

After completing this task, I developed an understanding of how logs can be used as evidence of past activities. I learned how timestamps and event details help establish timelines, recognise suspicious patterns, and support security investigations. This knowledge is useful for SOC monitoring, threat detection, and incident response.

Conclusion

Task 2 demonstrated the importance of logs as historical evidence in cybersecurity. By reviewing recorded events and arranging them in chronological order, security analysts can better understand system activities and investigate potential incidents. The task reinforced the value of accurate timestamps, proper log preservation, and careful analysis before reaching conclusions about suspicious behaviour.



Answer the questions below

Based on the list of log types in this task, what log type is used by the log file specified in the note from Task 2?

Web Server Log

Check

Based on the list of log formats in this task, what log format is used by the log file specified in the note from Task 2?

Combined

✓ Correct Answer

Task 3: Types, Formats, and Standards

Introduction

Logs are generated by different systems and applications, and each log source may record different types of information. Understanding log types, formats, and standards is important for analysing events accurately. Consistent log structures make it easier to search, filter, interpret, and compare records from different sources.

Task Description

This task focused on identifying different types of logs and understanding the formats used to represent log data. It introduced the importance of standardised log structures and explained how log formats help security analysts interpret events generated by servers, applications, and other systems.

Types of Logs

Different systems generate logs for specific purposes. Common examples include:

- **System Logs:** Record operating system events, warnings, and errors.
- **Authentication Logs:** Record login attempts and authentication-related activities.

- **Application Logs:** Provide information about application operations, errors, and performance.
- **Web Server Logs:** Record HTTP requests and responses involving web servers.
- **Security Logs:** Record security-related events that may help identify suspicious behaviour.

Each log type provides a different perspective on system activity and may be useful during an investigation.

Log Formats

A log format determines how information is organised within a log entry. Some logs use plain text, while others use structured formats that can be processed by monitoring tools.

The Combined Log Format is commonly used by web servers. It can record the client IP address, timestamp, requested resource, HTTP response status, and additional request information. Understanding the format helps analysts interpret the recorded activity correctly.

Logging Standards

Logging standards and consistent formatting practices help different systems and monitoring tools exchange and interpret information more effectively. Structured formats reduce ambiguity and support automated parsing, filtering, and analysis.

Consistent timestamps, field names, and event structures are particularly useful when investigating activities across multiple systems.

Importance in Cybersecurity

Correctly identifying a log type and its format helps analysts select relevant evidence and avoid misinterpreting records. For example, authentication logs are useful when investigating login attempts, while web server logs can help investigate unusual website requests.

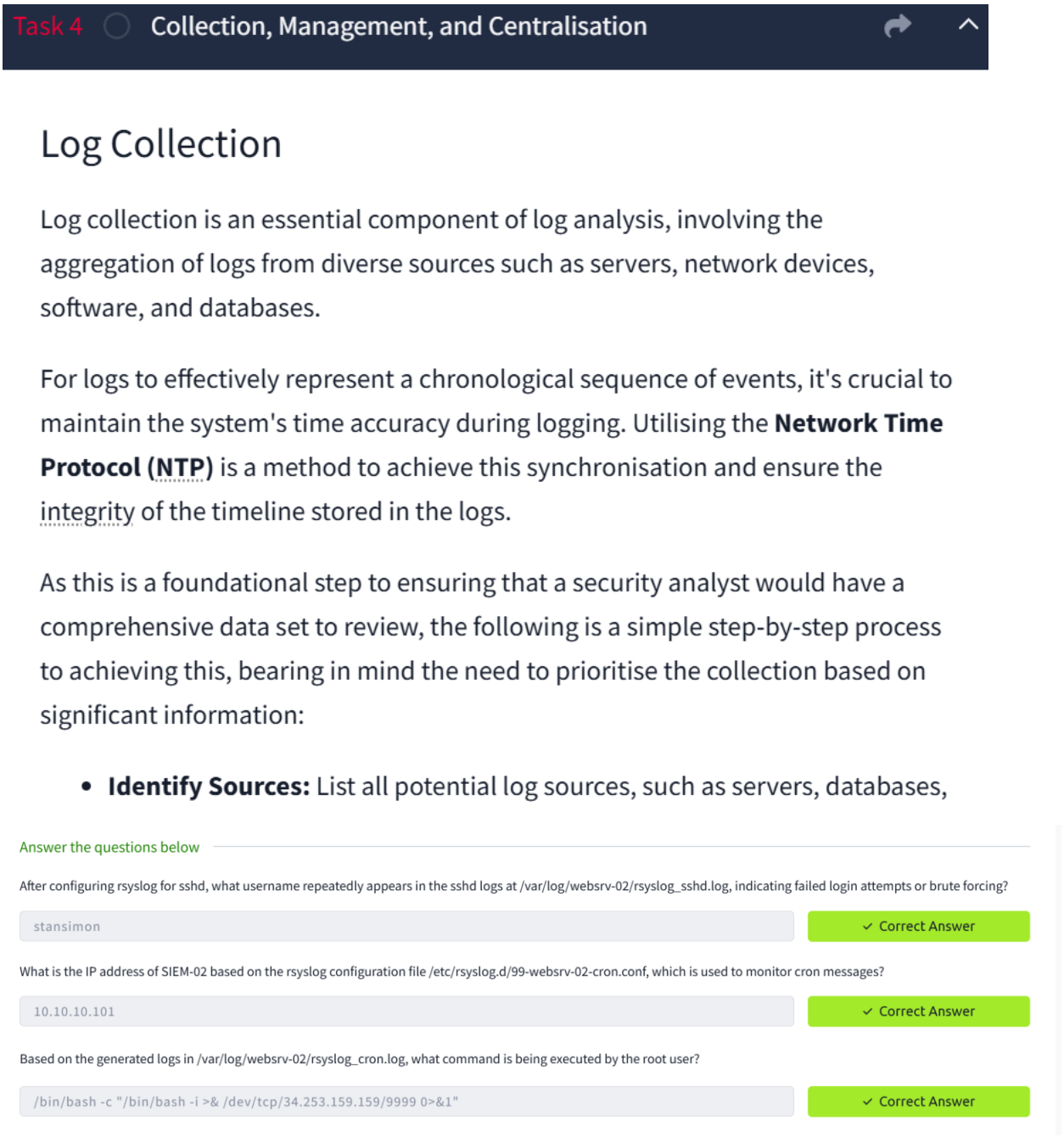
Standardised logs also improve the effectiveness of centralised monitoring and SIEM platforms.

Learning Outcomes

After completing this task, I gained an understanding of common log categories, log formats, and the importance of consistent logging practices. I learned how different log sources provide different information and how standardised formats support more effective analysis.

Conclusion

Task 3 demonstrated that understanding log types, formats, and standards is essential for reliable log analysis. Recognising the source and structure of a log helps security analysts interpret events accurately, investigate suspicious behaviour, and use monitoring tools more effectively.



Task 4: Collection, Management, and Centralisation

Introduction

Modern IT environments contain multiple systems that generate logs continuously. Managing these records efficiently is important for maintaining visibility and identifying potential security issues. Log collection and centralisation allow organisations to gather information from different sources and review it in a more organised way.

Task Description

This task focused on the methods used to collect, manage, and centralise log data. It introduced the role of logging services such as rsyslog in receiving, processing, and forwarding log messages. Centralised logging makes it easier for security analysts to investigate events without accessing each system individually.

Log Collection

Log collection involves gathering event records from operating systems, applications, servers, and network devices. The collected information can help administrators monitor system health, investigate errors, and detect unusual activities.

Reliable collection is important because missing or incomplete logs can make investigations more difficult.

Log Management

Log management involves organising, processing, storing, and maintaining log records throughout their lifecycle. Proper management helps ensure that logs remain accessible, useful, and suitable for analysis.

It also involves controlling access to sensitive records and maintaining appropriate policies for storage and retention.

Centralised Logging

Centralised logging brings log data from multiple systems into a common destination. This approach improves visibility, simplifies searching, and allows analysts to compare events from different sources.

A centralised logging system can help identify patterns that may not be obvious when each log file is examined separately.

Understanding rsyslog

Rsyslog is a logging service that can receive, process, and forward log messages according to its configuration. It is commonly used in Linux environments to manage system logs and transfer records to remote logging servers.

In the practical exercises, rsyslog configuration files were examined to understand how logs could be forwarded to a central destination. The SIEM-02 IP address identified in the lab configuration was 10.10.10.101.

Importance in Cybersecurity

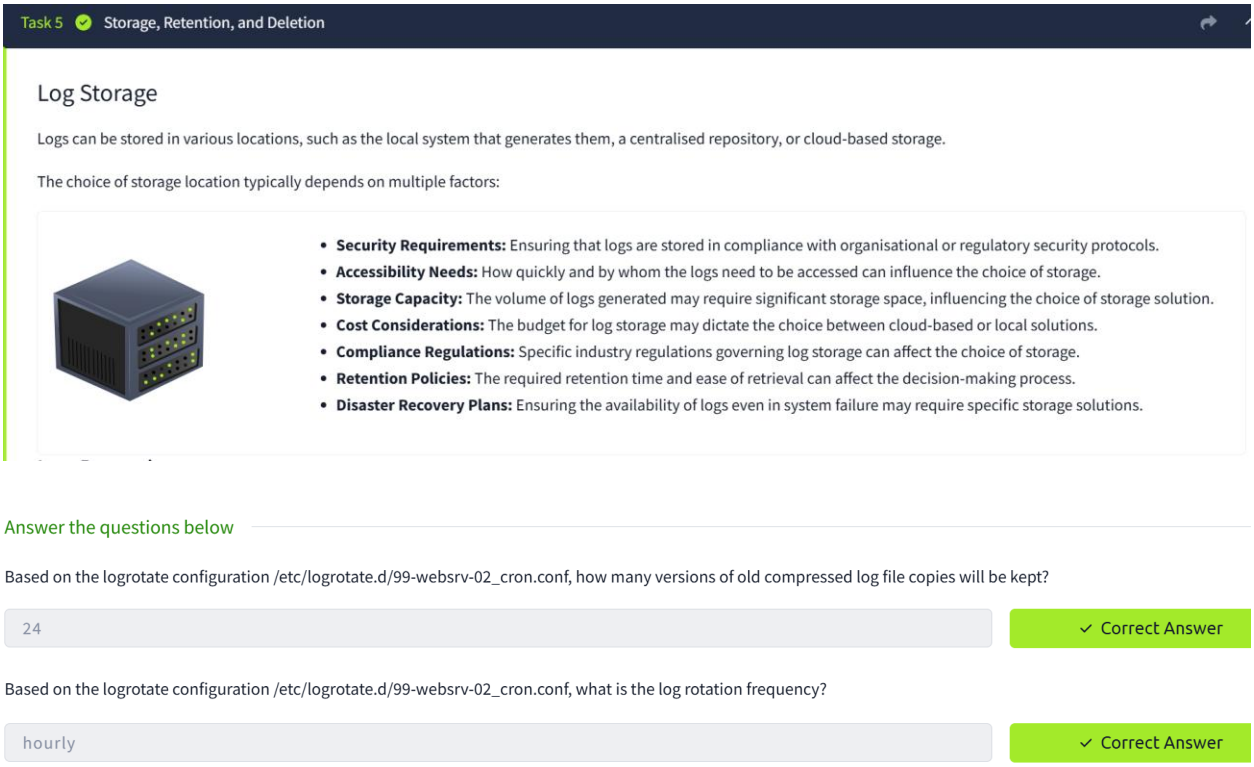
Centralised logging supports security monitoring by bringing together information from different sources. Analysts can use these records to investigate suspicious authentication attempts, review system activity, and identify events that require further examination.

Learning Outcomes

After completing this task, I gained an understanding of log collection, management, and centralisation. I also learned about the purpose of rsyslog and how configuration files can help identify the destination used for forwarding log messages.

Conclusion

Task 4 demonstrated the importance of collecting and centralising logs for effective system monitoring and security investigations. Centralised logging improves visibility across an IT environment and helps analysts review related events more efficiently.



Task 5: Storage, Retention, and Deletion

Introduction

Log files can consume considerable storage space because systems continuously generate event records. Organisations therefore need suitable policies for storing, rotating, retaining, and deleting logs. These practices help maintain system performance while preserving records required for investigations and compliance.

Task Description

This task explored how logs are stored and managed over time. It focused on log rotation, retention periods, compression, and deletion. The task demonstrated why log management

policies must balance storage requirements with the need to preserve useful historical information.

Log Storage

Log storage involves keeping event records in files or logging systems so that they can be accessed when required. Storage arrangements should support efficient searching and protect records against unauthorised access or accidental loss.

Important security logs may need additional protection because they can contain sensitive information about users, systems, and network activities.

Log Rotation

Log rotation prevents individual log files from growing indefinitely. When a rotation condition is met, a new log file may be created while the previous file is retained under a different name.

Rotated files may also be compressed to reduce their storage requirements. Rotation helps keep log management organised and reduces the risk of excessive disk usage.

Log Retention

Log retention defines how long records should be kept before they are deleted or archived. The appropriate retention period depends on operational needs, investigation requirements, organisational policies, and applicable regulations.

Keeping logs for a suitable period allows analysts to investigate historical events, while excessive retention may increase storage costs and privacy risks.

Log Deletion

Log deletion removes records that are no longer required under the applicable retention policy. Deletion should be managed carefully so that important evidence is not removed prematurely.

Organisations should ensure that log deletion follows approved policies and that required records remain available for investigations or audits.

Practical Findings

During the lab exercise, the cron log rotation configuration was examined. The observed configuration specified hourly rotation and retention of 24 rotated copies.

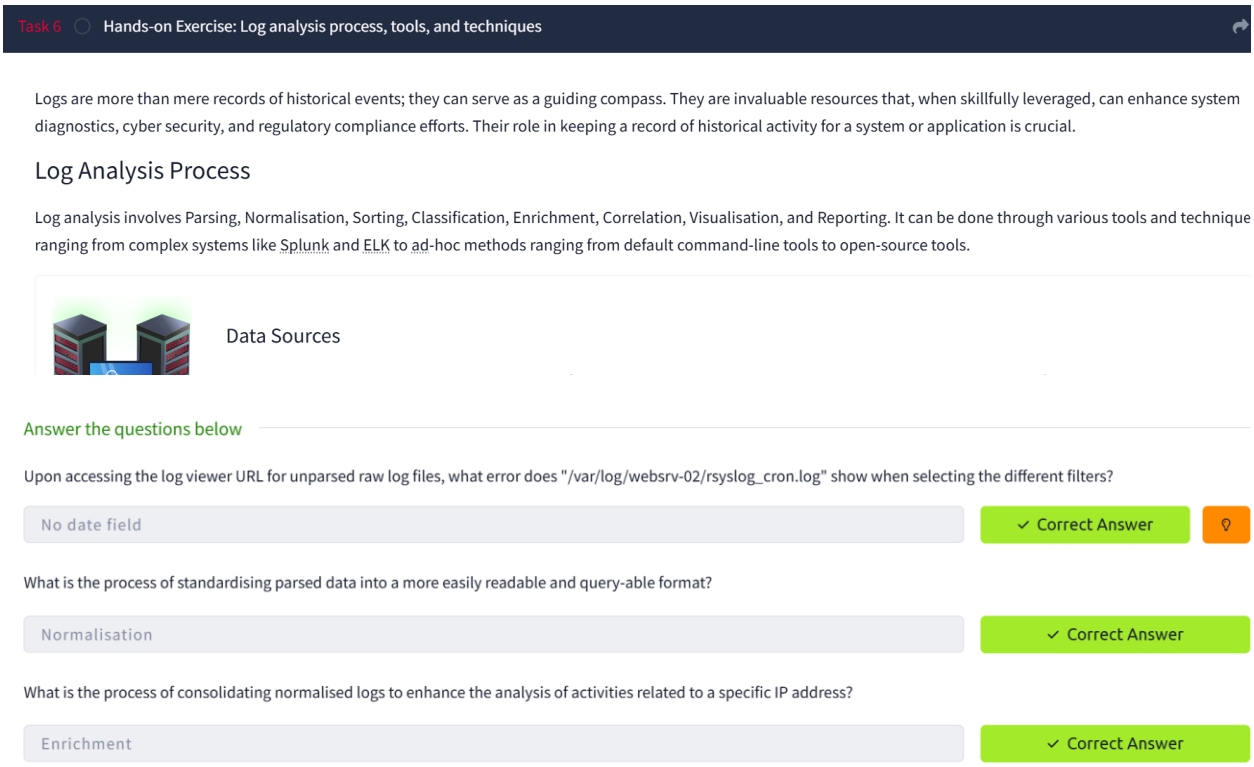
These settings illustrated how rotation frequency and retention limits can be configured to manage the number of stored log files. Actual storage requirements and deletion behaviour depend on the complete configuration.

Learning Outcomes

After completing this task, I gained an understanding of log storage, rotation, retention, and deletion. I learned how rotation helps manage file sizes and how retention policies determine the period for which historical records remain available.

Conclusion

Task 5 demonstrated that effective log storage and retention are essential for reliable monitoring and investigation. Appropriate rotation and retention policies help manage storage resources while preserving useful evidence for troubleshooting, security investigations, and compliance.



Task 6: Hands-on Exercise — Log Analysis Process, Tools, and Techniques

Introduction

Practical log analysis is an important cybersecurity skill because logs provide evidence of activities occurring across systems and networks. Security analysts examine log records to identify unusual patterns, investigate authentication problems, and detect possible indicators of compromise.

Task Description

This task provided practical experience in examining log files, reviewing logging configurations, and interpreting events associated with web server requests, SSH authentication, and scheduled cron activities. It also introduced issues related to log parsing, filtering, normalisation, and enrichment.

Web Server Log Analysis

The lab examined the following web server log:

/var/log/gitlab/nginx/access.log

Web server access logs record requests made to a website and may include client addresses, timestamps, requested resources, HTTP status codes, and user-agent information.

Analysing these records helps investigators understand website activity and identify request patterns that may require further examination.

SSH Authentication Log Analysis

The SSH log examined during the practical exercise was:

/var/log/websrv-02/rsyslog_sshd.log

The username stansimon repeatedly appeared in the SSH records associated with failed login attempts. Repeated authentication failures can indicate password guessing or possible brute-force activity.

However, repeated failures alone do not establish that an attack succeeded. Additional records and context are needed to determine the nature and impact of the activity.

SIEM Configuration Analysis

The configuration file examined during the exercise was:

/etc/rsyslog.d/99-websrv-02-cron.conf

The SIEM-02 IP address identified in the lab configuration was:

10.10.10.101

This finding demonstrated how logging configuration files can provide information about the destination to which log messages are forwarded for centralised monitoring.

Cron Log Analysis

The cron log examined during the exercise was:

/var/log/websrv-02/rsyslog_cron.log

The log contained a suspicious command attempting to establish a reverse shell connection to 34.253.159.159 on port 9999.

A reverse shell attempt may allow a remote host to interact with a command shell on another system. Such a command is a significant security indicator and should be investigated carefully.

The presence of the command in a log does not, by itself, prove that the connection succeeded or that the system was compromised.

Log Parsing and Filtering

Log parsing converts raw log entries into fields that can be searched and analysed by tools. Filtering allows analysts to focus on records matching particular criteria, such as timestamps, event types, or IP addresses.

During the lab, the log viewer displayed the error No date field when different filters were selected for the cron log. This indicated that the expected date field was unavailable to the selected filtering process, preventing the relevant date-based operation from working as expected.

Normalisation and Enrichment

Normalisation standardises log data into a consistent structure so that records from different sources can be interpreted more easily.

Enrichment adds contextual information to existing log records, helping analysts understand the significance of events. These techniques improve the usefulness of log data and can support investigations involving specific IP addresses, users, or systems.

Key Findings

The practical exercises produced the following observations:

- The web server access log provided a source for reviewing historical HTTP requests.
- The username stansimon repeatedly appeared in SSH records associated with failed login attempts.
- The SIEM-02 address identified in the configuration was 10.10.10.101.
- The cron log contained a suspicious reverse-shell command targeting 34.253.159.159 on port 9999.
- The log viewer displayed a No date field error during filtering.

These observations illustrate how different log sources can contribute to an investigation. They should be treated as lab findings requiring appropriate interpretation rather than automatic proof of a successful compromise.

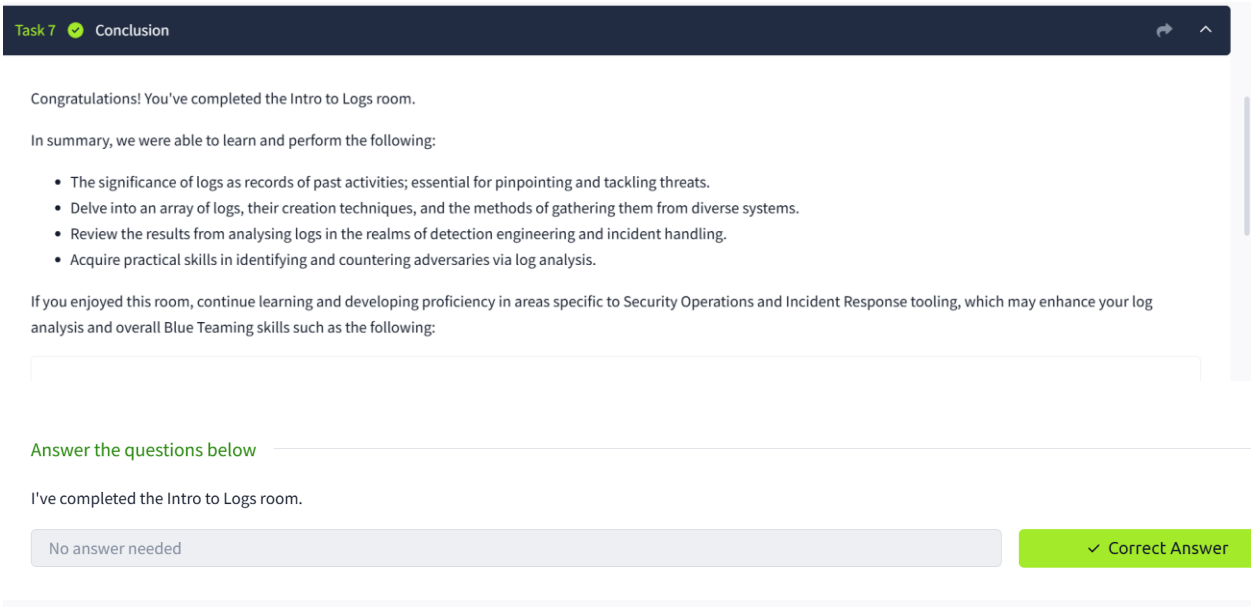
Learning Outcomes

After completing this task, I gained practical exposure to examining web server logs, reviewing SSH authentication records, inspecting rsyslog configuration, and identifying suspicious

commands in cron logs. I also learned how parsing errors can affect filtering and how normalisation and enrichment improve log analysis.

Conclusion

Task 6 demonstrated how practical log analysis can support security monitoring and incident investigation. Examining multiple log sources helped illustrate how authentication failures, unusual commands, and configuration details can provide useful investigative leads. The task also reinforced the importance of accurate parsing, contextual analysis, and verification before drawing conclusions about suspicious activity.



Task 7: Conclusion

Introduction

The final task brought together the concepts covered throughout the Intro to Logs room. Logging supports system monitoring, troubleshooting, security investigations, and incident response by maintaining records of activities occurring across an IT environment.

Task Description

This task concluded the learning process by reinforcing the importance of understanding log sources, formats, collection methods, storage policies, and analysis techniques. It connected the theoretical concepts introduced in the earlier tasks with the practical observations made during the hands-on exercise.

Summary of Learning

Throughout the room, I learned that logs provide historical evidence of system and application activities. I explored different log types and formats and learned why consistent structures are useful for analysis.

The room also introduced centralised logging, rsyslog, log rotation, retention policies, parsing, normalisation, and enrichment. The practical exercise demonstrated how web server, SSH, and cron logs can provide information relevant to security investigations.

Importance of Log Analysis in Cybersecurity

Log analysis is an important activity in Security Operations Centers because it helps analysts monitor systems and identify events that require further investigation. Reviewing authentication records, server activity, and system commands can reveal patterns that might otherwise go unnoticed.

Effective analysis depends on accurate timestamps, reliable collection, appropriate retention, and careful interpretation of the evidence.

Learning Outcomes

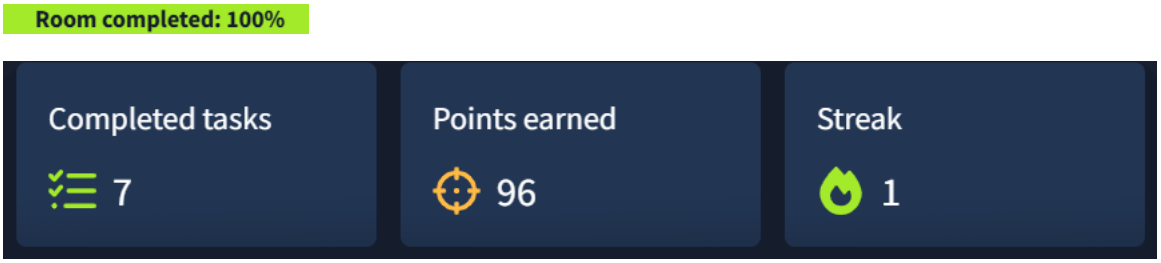
After completing all seven tasks, I developed a foundational understanding of logging and its role in cybersecurity. I gained knowledge of log collection and management and practical exposure to examining log files and identifying potentially suspicious activity.

This learning provides a foundation for further study in SOC operations, SIEM monitoring, threat detection, and incident response.

Conclusion

The TryHackMe Intro to Logs room provided a structured introduction to the role of logs in cybersecurity. The tasks demonstrated how log records can support historical investigations, how centralised collection improves visibility, and how storage and retention policies help preserve useful evidence.

The practical exercises strengthened my understanding of web server logs, SSH authentication records, cron logs, rsyslog configuration, and common log-analysis challenges. Overall, the room established a useful foundation for further learning in security monitoring and incident investigation.



Five Types of Logs and Their Sources

Logs are records of events generated by computers, applications, servers, and network devices. They help security analysts monitor activity, troubleshoot problems, and identify suspicious behaviour.

Log Type	Source	Purpose
System Logs	Operating system and system services	Record system startup, shutdown, errors, and service activity.
Security Logs	Windows Security, Linux authentication services, and login systems	Record login attempts, authentication events, and access-related activities.
Application Logs	Software, applications, and databases	Record application errors, warnings, transactions, and operational events.
Network Logs	Firewalls, routers, VPNs, and intrusion detection systems	Record network connections, traffic, blocked requests, and communication events.
Web Server Logs	Apache and Nginx web servers	Record website requests, client IP addresses, requested resources, and HTTP status codes.

Apache Web Server Log Analysis

Apache web server logs help administrators and security analysts understand how users interact with a website. They can also help identify unusual requests, repeated errors, and potentially suspicious activity.

Example Apache log entry:

203.0.113.24 - - [27/Oct/2026:10:15:32 +0000] "GET /index.html HTTP/1.1" 200 512 "-" "Mozilla/5.0"

Field	Value	Explanation
IP Address	`203.0.113.24`	Identifies the client address recorded in the example.
Timestamp	`27/Oct/2026:10:15:32 +0000`	Shows when the request occurred, including the UTC offset.
Request	`GET /index.html HTTP/1.1`	Shows that the client requested the `/index.html` resource using HTTP GET.
Status Code	`200`	Indicates that the server successfully handled the request.
Response Size	`512`	Indicates the response size in bytes, excluding headers in this example format.
Referrer	`-`	Indicates that no referrer was recorded.
User-Agent	`Mozilla/5.0`	Identifies the client software string recorded in the request.

Normal vs. Suspicious Log Entries

Normal Log Entry

203.0.113.24 - - [27/Oct/2026:10:15:32 +0000] "GET /index.html HTTP/1.1" 200 512

Analysis: This entry shows a successful request for a normal website page. If the request matches the website's expected activity, it may be considered routine. However, context is always important when evaluating logs.

Potentially Suspicious Log Entry

198.51.100.77 - - [27/Oct/2026:10:18:09 +0000] "GET /wp-admin HTTP/1.1" 401 178

Analysis: This entry shows a request to /wp-admin that received HTTP status code 401, indicating that authentication was required or unsuccessful. Such a request may deserve investigation if it is unexpected or part of repeated attempts to access restricted resources.

A single suspicious-looking entry does not prove that an attack occurred. Analysts should examine related requests, authentication records, timestamps, and other evidence before reaching a conclusion.

Baseline in Log Analysis

A baseline is a reference representation of normal activity within a system, application, or network. Security analysts use it to compare current log activity with expected behaviour and identify unusual changes.

For example, a website may normally receive a predictable number of requests during business hours. If it suddenly receives a large number of requests to restricted pages, this activity may differ from its established baseline and require investigation.

A useful baseline may include:

- Typical login times and authentication patterns.
- Normal network traffic volume and connection frequency.
- Common website URLs and HTTP status codes.
- Expected system events and application activity.
- Usual user accounts, devices, and service behaviour.

Baselines should be developed using representative data and updated when normal operations change. An anomaly is not automatically malicious; it is a signal that may need further investigation.

Basic Log Analysis Process

A structured log analysis process helps analysts investigate events consistently:

1. **Collect logs:** Gather relevant records from operating systems, applications, servers, and network devices.
2. **Validate the data:** Check timestamps, log formats, and important fields for missing or inconsistent information.
3. **Filter and search:** Focus on relevant time periods, IP addresses, user accounts, event IDs, and status codes.
4. **Identify unusual activity:** Look for repeated failed logins, unexpected access attempts, unusual traffic, or sudden changes in event frequency.
5. **Compare with the baseline:** Determine whether the activity differs from normal behaviour.
6. **Correlate evidence:** Compare related logs to understand what happened and whether the activity is legitimate or suspicious.

7. **Document findings:** Record the evidence, observations, investigation steps, and recommended actions.

Conclusion

Log analysis is an important part of cybersecurity because it allows analysts to understand system activity, investigate security events, and detect potentially suspicious behaviour. Learning about different log types and their sources provides a foundation for effective monitoring. Analysing Apache access logs helps identify client requests, timestamps, response codes, and unusual access patterns. Comparing events with a baseline helps analysts recognise deviations from normal activity, while additional evidence helps determine whether those deviations represent a genuine security concern. The TryHackMe learning activities supported the understanding of log evidence and Windows investigation concepts. Together, these skills provide a foundation for further work in Security Operations Centre (SOC) monitoring and incident investigation.